COASTAL AIR ASSESSMENTS

Home Environmental Control Plan

How to keep your home dry, clean, and low in dust, explained with building science and written so everyone in the family can follow along.

THE SIMPLE VERSION

Tiny living things like mold and dust mites need water to grow.
If we keep the house dry and clean, they cannot grow very much, and people breathe easier.

The goal of this plan

When someone in the home has mold or dust sensitivities, the building should be making things better, not worse. This plan lowers the ways a house can add to the problem, using four layers in order of impact: **control moisture**, **filter the air**, **remove dust reservoirs** (bedrooms first), and **clean in a way that removes allergens instead of stirring them up**. If symptoms continue once these are in place, the home is far less likely to be the main driver. Allergists recommend this combined approach because single steps used alone rarely bring a clear improvement in symptoms.[1]

How to read this guide. Every section opens with a yellow **Simple version** box written in short, plain sentences that a young reader can follow. The blue boxes and tables below it give the science and the exact steps for the adults in the home.

Where to start

If you can only do five things this month, do these. They give the largest return for the effort.

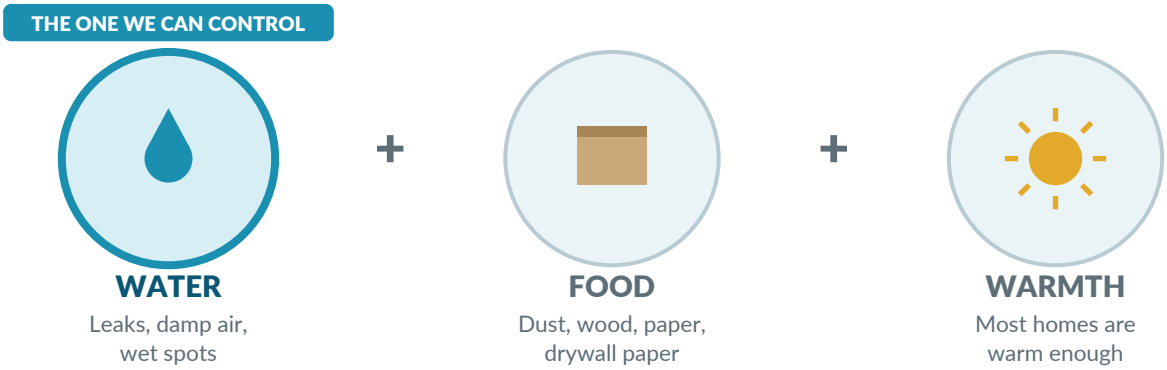
#	Action	Why it matters
1	Keep indoor humidity between 35 and 50 percent, and fix anything that keeps it above 50 percent for long periods. Check it with a hygrometer in the bedroom and main living area.	Allergists recommend 35 to 50 percent to limit dust mites.[1] EPA advises below 60 percent, ideally 30 to 50 percent, to limit mold.[2]
2	Fix leaks, water intrusion, and condensation, and dry any wet material within 24 to 48 hours.	Cleaning cannot keep up with an active water source. Most materials dried within 24 to 48 hours will not grow mold.[2]
3	Put zippered allergen-barrier covers on the mattress, box spring, and pillows, and wash bedding every week.	Encasings are a strong recommendation in the allergy practice parameter, and weekly washing removes most mites and allergen.[1]
4	Set the thermostat fan to AUTO, and use a MERV 13 filter (or the highest your system can handle) that fits with no gaps.	Fan ON puts coil water back into the house.[3] EPA recommends at least MERV 13 or as high as the system allows.[4]
5	Run a correctly sized true-HEPA air cleaner in the sensitive person's bedroom.	HEPA cleaners that filter the breathing zone during sleep appear to be the most helpful type of filtration.[1]

PART 1 OF 11 · START HERE

How mold and dust mites work

THE SIMPLE VERSION

Mold is a tiny living thing. It needs water, food, and warmth.
Our homes always have food (dust) and warmth, so we take away the water.
Dust mites are tiny bugs, too small to see. They drink water out of the air.
If the air stays dry, mold and dust mites cannot grow very much.



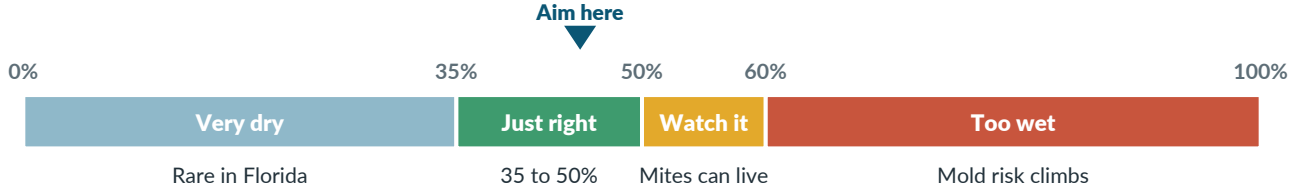
Mold needs moisture, a food source, and a suitable temperature. Food and warmth are present in every home, so water is the control point.

Mold

Mold spores are everywhere, indoors and out, so the aim is not a spore-free house. The aim is to keep materials dry enough that spores never get the moisture they need to grow. What matters most is the moisture **at the surface of a material**, not just the humidity in the middle of the room. Building scientists find that mold growth is rarely a problem when material surfaces stay below about 80 percent relative humidity on a 30-day average, the criterion used in ASHRAE Standard 160.[5] A wall, closet, or AC vent that runs cooler than the room can reach that level even when the hygrometer in the room looks fine.

Dust mites

Dust mites are tiny bugs that live in bedding, upholstery, and carpet, where they eat skin flakes. They do not drink. They soak up water from the air, which is why humidity controls them so well. When relative humidity stays below about 50 percent, the moisture inside bedding and upholstered furniture falls below what mites need to survive.[1] Short humid periods still count: mites can survive if humidity goes above 50 percent for as little as one to one and a half hours a day, and two to three hours a day lets them reproduce.[1] That is why showers, cooking, and AC settings matter every day, not just on average.



Your indoor humidity target. Allergists recommend 35 to 50 percent[1]; EPA calls 30 to 50 percent ideal and advises below 60 percent.[2]

Your next step

Start with the five actions on page 1. Your assessment report shows which parts of this plan matter most for your home, and each part is a short guide of its own.

Sources for this part

1. AAAAI/ACAAI Joint Task Force, Environmental Assessment and Exposure Control of Dust Mites: A Practice Parameter (2013). https://www.aaaai.org/Aaaai/media/Media-Library-PDFs/Allergist%20Resources/Statements%20and%20Practice%20Parameters/Dust mite-2013.pdf
2. EPA, A Brief Guide to Mold, Moisture and Your Home. https://www.epa.gov/mold/brief-guide-mold-moisture-and-your-home
3. Florida Solar Energy Center (UCF), Mold Growth: How to Minimize Mold Growth. https://energyresearch.ucf.edu/consumer/buildings/building-science-basics/mold-growth/
4. EPA, Guide to Air Cleaners in the Home. https://www.epa.gov/indoor-air-quality-iaq/guide-air-cleaners-home
5. ASHRAE Handbook, Moisture Management in Buildings (Harriman and Lstiburek; ASHRAE Standard 160 criteria). https://handbook.ashrae.org/Handbooks/A15/SI/a15_ch62/a15_ch62_si.aspx